

LAB EXPERIMENT RECORD

08-10-2026---Set 13: Geographic Data Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	08 OCTOBER 2026
Day	Thursday

Aim

To analyze geographic data of three cities using a map chart, scatter plot and table, and to build an interactive Tableau dashboard for exploring the geographic data.

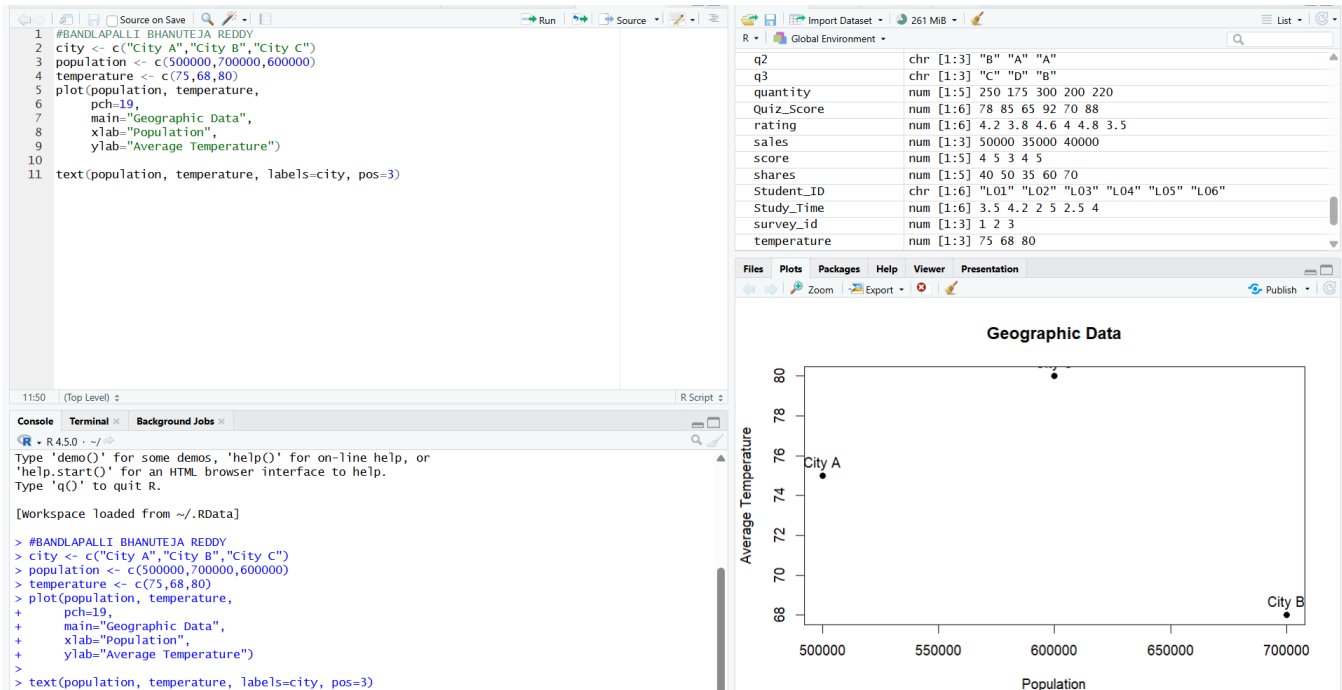
Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Question 1: Map Chart of Cities

Question: Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Aim: To show the three cities on a geographic map using labelled markers sized by population.

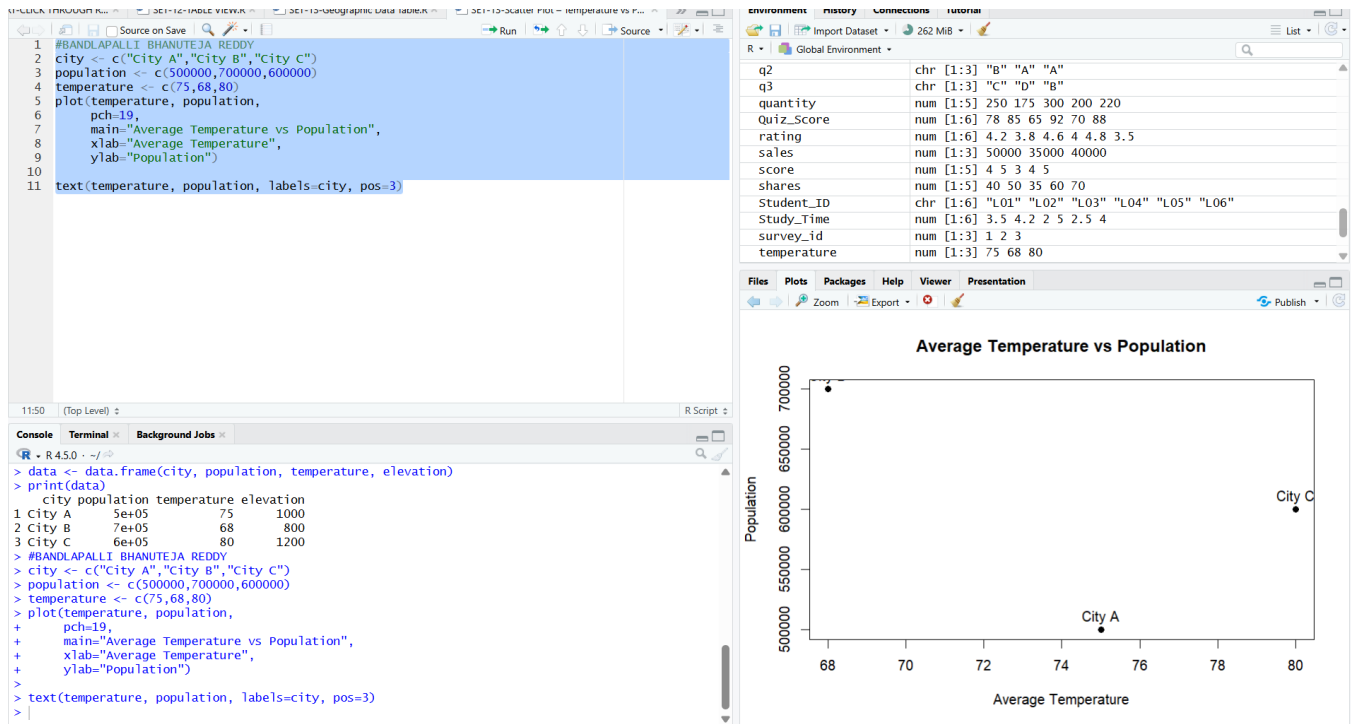


Result: City B has the largest marker (700000), followed by City C (600000) and City A (500000); the dataset has no latitude or longitude, so the real locations must be supplied.

Question 2: Scatter Plot of Temperature vs Population

Question: Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

Aim: To explore how average temperature and population size are related using a scatter plot.



Result: The correlation is about -0.58, a moderate negative relationship, because the coolest city (City B, 68) has the highest population; with only three cities this is only an indication.

Question 3: Table of Geographic Data

Question: Build a table to display the geographic data for each city. Label the table elements.

Aim: To present the population, average temperature and elevation of each city in a clearly labelled table.

The screenshot shows the R Studio environment. The script editor contains the following R code:

```

1 #BANDLAPALLI BHANUTEJA REDDY
2 city <- c("City A","City B","City C")
3 population <- c(500000,700000,600000)
4 temperature <- c(75,68,80)
5 elevation <- c(1000,800,1200)
6 data <- data.frame(city, population, temperature, elevation)
7 print(data)

```

The console output shows the execution of the code, including the creation of the data frame and the resulting table:

```

+ pch=19,
+ main="Average Temperature vs Population",
+ xlab="Average Temperature",
+ ylab="Population")
>
> text(temperature, population, labels=city, pos=3)
> #BANDLAPALLI BHANUTEJA REDDY
> city <- c("City A","City B","City C")
> population <- c(500000,700000,600000)
> temperature <- c(75,68,80)
> elevation <- c(1000,800,1200)
> data <- data.frame(city, population, temperature, elevation)
> print(data)
  city population temperature elevation
1 City A      5e+05         75      1000
2 City B      7e+05         68         800
3 City C      6e+05         80      1200
>

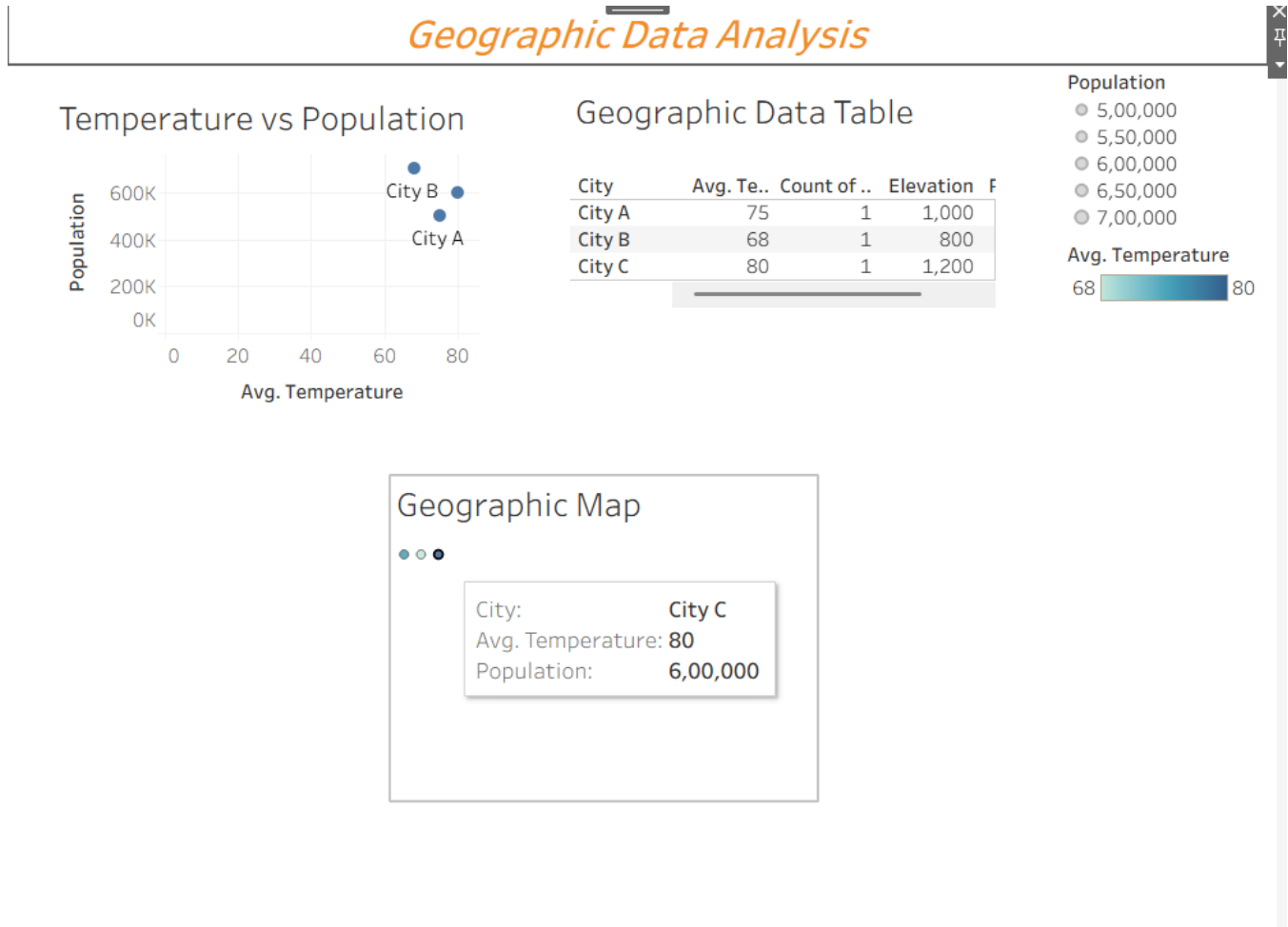
```

Result: The table lists all three cities; City C is the hottest (80) and the highest (1200), while City B has the largest population and the lowest elevation (800).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Aim: To build an interactive Tableau dashboard that combines the map chart, scatter plot and table for exploring geographic data.



Result: The dashboard places all three views together, and selecting a city on the map highlights it in the scatter plot and the table.

Overall Result

Observation: The three cities have a total population of 1,800,000, an average temperature of about 74.3 and an average elevation of 1000.